INCH-POUND
MIL-PRF-3098/42K
13 April 2010
SUPERSEDING
MIL-PRF-3098/42J
9 December 2004

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR64/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2.4 MHz to 20 MHz; fundamental; noncontrolled; anti-resonance.

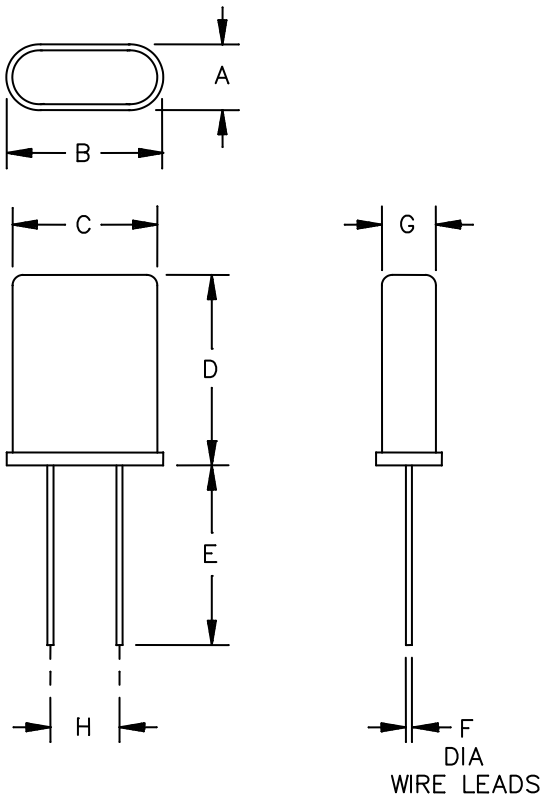


FIGURE 1. Crystal unit - CR64/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.183	---	4.65
B	---	.435	---	11.05
C	---	.402	---	10.21
D	---	.`530	---	13.46
E	.500	---	12.70	---
F	.016	.019	0.41	0.48
G	---	.150	---	3.81
H	.184	.200	4.67	5.08

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR64/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.4 MHz to 20 MHz, inclusive.

Frequency tolerance:

Operating temperature range: ±50 parts per million (ppm).

Antiresesonace, load capacitance: 30.0 pF ±0.5 pF.

Equivalent resistance: See table I.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Capacitance, shunt: 2.4 MHz to 4.0 MHz - 1.7 pF min. to 7.0 pF max.
4.0 MHz to 6.0 MHz - 2.7 pF min. to 7.0 pF max.
6.0 MHz to 20.0 MHz - 3.7 pF min. to 7.0 pF max.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

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TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
2.40 to 2.60	300
2.61+ to 2.89	250
2.90+ to 3.75	180
3.75+ to 4.75	120
4.75+ to 6.00	75
6.00+ to 7.00	50
7.00+ to 10.00	30
10.00+ to 20.00	25

Part or Identifying Number (PIN): CR64/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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